

Combinatorial covering designs

OPG-762

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

A (v, k, t) covering design, or covering, is a family of k -subsets, called blocks, chosen from a v -set, such that each t -subset is contained in at least one of the blocks. The number of blocks is the covering's size, and the minimum size of such a covering is denoted by $C(v, k, t)$.

Problem Find a closed form, recurrence, or better bounds for $C(v, k, t)$. Find a procedure for constructing minimal coverings.